BCSAI 704: R PROGRAMMING LANGUAGE

2L + 0T + 0P + 2C **MM 100**

Unit 1: Introduction to R Environment

History and development of R Statistical computing programming language, installing R and R studio, getting started with R, creating new working directory, changing existing working directory, understanding the different data types, installing the available packages, calling the installed packages, arithmetic operations, variable definition in R, simple functions, vector definition and logical expressions, matrix calculation and manipulation using matrix data types, workspace management.

Unit 2: Data Structures, Looping and Branching

Introduction to different data types, vectors, atomic vectors, types and tests, coercion, lists, list indexing, function applying on the lists, adding and deleting the elements of lists, attributes, name and factors, matrices and arrays, matrix indexing, filtering on matrix, generating a covariance matrix, applying function to row and column of the matrix, data frame – creating, coercion, combining data frames, special types in data frames, applying functions: lapply( ) and sapply( ) on data frames, control statements, loops, looping over non vector sets, arithmetic and Boolean operators and values, branching with **if**, looping with **for**, **if-else** control structure, looping with **while**, vector based programming.

Unit 3: R - Object Oriented Programming

Introduction to object oriented concepts in R, basics of S3 classes – S3 Generic functions, OPP in linear model functions, writing S3 classes, using inheritance, introduction to S4 classes, writing S4 Classes, implementing a generic function on an S4 Classes, comparison of S3 and S4 classes, management of objects – listing objects, removing specific objects from the existing function and working directory, saving the collection of objects with save( ) function.

Unit 4: R for Statistics

Descriptive statistics – mean (arithmetic, geometric and harmonic), median, mode for raw and grouped data, measure of dispersion – range, standard deviation, variance, coefficient of variation, testing of hypothesis – small sample test, large sample test – for comparing mean, proportion, variance, correlation and regression – significance of correlation and regression coefficients, chi-square test, non-parametric test, Analysis of Variance for one way variation and two variation – with and without interaction.

Unit 5: R with C, C++ and Python

Introduction to C and C++ programming concepts, writing C/C++ functions to be called from R, preliminaries of R to C and C++ programming languages, some mathematical programming example with R and C/C++, compiling and running the code, debugging R/C code, introduction to Python and its components, installing packages related with python in R, syntax of RPy packages.

Text Book

1. The art of R programming – Norman Matloff, no starch Press, San Francisco.
2. R in Action – Robert I. Kabacoff, Second Edition, Dreamtech Press.
3. Introduction to Scientific Programming and Simulation using R – Owen Jones, Robert Maillardet and Andrew Robinson, CRC Press
4. Advanced R – Hadley Wickham, CRC Press.